

Introduction to mathematics for economists

Exercise Sheet #3

Alexandre Grellet

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During the tutorials, we will solve indefinite integrals 8 and 11, definite integrals 1 and 3, the second-price auction exercise, the waiting-time exercise, and the wage-bargaining exercise. Prioritise these. The **discrete lottery** exercise is a recommended warm-up before the continuous probability exercises. A full answer sheet is provided, since we will not have time to solve everything in class.

Indefinite Integrals

Indefinite Integral

By **indefinite integral**, we mean an antiderivative (or primitive) of the function f . For instance, the antiderivative of $f(x) = x$ is:

$$\int x \, dx = \frac{x^2}{2} + C$$

where C is just a constant. We call it *indefinite* because we do not specify the bounds — we only look for the primitive.

Compute the following indefinite integrals (do not forget to specify the definition set). Hints are given for the ones that need a technique beyond the standard table of antiderivatives.

1. $\int x^\alpha \, dx$, with $\alpha \in \mathbb{R}$.
2. $\int x^2 e^{-x} \, dx$.
3. $\int x \ln(x) \, dx$.
4. $\int \cos^2(x) \, dx$.
5. $\int \frac{x}{x^2 + 1} \, dx$.
6. $\int \frac{1}{x^2 - 1} \, dx$. *Hint: you may use $\frac{1}{x^2 - 1} = \frac{1/2}{x - 1} - \frac{1/2}{x + 1}$*
7. $\int (ax + b)^n \, dx$, with $a \neq 0$, $b \in \mathbb{R}$, $n \in \mathbb{N}$.
8. $\int \frac{\ln(x)}{x} \, dx$.
9. $\int \frac{\ln^n(x)}{x} \, dx$, with $n \in \mathbb{N}$.

10. $\int \frac{2x+1}{x^2+x+1} dx$.
11. $\int \frac{1}{1+x^2} dx$. (Hint: recall that $\frac{d}{dx} \arctan x = \frac{1}{1+x^2}$.)

Solution

1. $\int x^\alpha dx$ is defined for $x \neq 0$ whenever $\alpha < 0$ and $\forall x$ otherwise. If $\alpha \neq -1$:

$$\int x^\alpha dx = \frac{x^{\alpha+1}}{\alpha+1} + C.$$

If $\alpha = -1$:

$$\int x^{-1} dx = \ln(x) + C.$$

2. Defined on $\mathbb{R}$, double integration by parts:

$$\int x^2 e^{-x} dx = -(x^2 + 2x + 2)e^{-x} + C.$$

3. Defined for $x > 0$, integration by parts ($u = \ln x$, $dv = x dx$):

$$\int x \ln(x) dx = \frac{x^2}{2} \ln(x) - \frac{x^2}{4} + C.$$

4. Defined on $\mathbb{R}$, using $\cos^2(x) = \frac{1+\cos(2x)}{2}$:

$$\int \cos^2(x) dx = \frac{x}{2} + \frac{\sin(2x)}{4} + C.$$

5. Defined on $\mathbb{R}$. Substitution $u = x^2 + 1$, $du = 2x dx$:

$$\int \frac{x}{x^2+1} dx = \frac{1}{2} \ln(x^2+1) + C.$$

6. Defined on $(-\infty, -1)$, $(-1, 1)$, or $(1, \infty)$. Partial fractions $\frac{1}{x^2-1} = \frac{1/2}{x-1} - \frac{1/2}{x+1}$ give

$$\int \frac{1}{x^2-1} dx = \frac{1}{2} \ln \left| \frac{x-1}{x+1} \right| + C.$$

Note: the constant could also differ on the different subintervals $(-\infty, -1)$, $(-1, 1)$, or $(1, \infty)$

7. Defined on $\mathbb{R}$. Substitution $u = ax + b$, $du = a dx$:

$$\int (ax+b)^n dx = \frac{(ax+b)^{n+1}}{a(n+1)} + C.$$

8. Defined for $x > 0$. Substitution $u = \ln(x)$, $du = dx/x$:

$$\int \frac{\ln(x)}{x} dx = \frac{1}{2} [\ln(x)]^2 + C.$$

9. For $x > 0$, same substitution:

$$\int \frac{\ln^n(x)}{x} dx = \frac{\ln^{n+1}(x)}{n+1} + C.$$

10. Defined on $\mathbb{R}$ (the denominator $x^2 + x + 1 > 0$ has no real root). Substitution $u = x^2 + x + 1$, $du = (2x + 1) dx$:

$$\int \frac{2x + 1}{x^2 + x + 1} dx = \ln(x^2 + x + 1) + C.$$

11. Defined on $\mathbb{R}$:

$$\int \frac{1}{1 + x^2} dx = \arctan(x) + C.$$

Definite and Improper Integrals

Compute the following integrals, when they are well defined.

1. $\int_1^\infty \frac{1}{x^2} dx.$
2. $\int_1^\infty \frac{1}{x} dx.$
3. $\int_0^\infty e^{-x} dx.$
4. $\int_e^\infty \frac{1}{x(\ln x)^2} dx.$ (*Hint: substitution $u = \ln x.$*)
5. $\int_{-1}^1 \frac{1}{x} dx$

Solution

We fix $A > 0$ and compute:

1. $\int_1^A x^{-2} dx = [-x^{-1}]_1^A = 1 - \frac{1}{A} \rightarrow 1$ as $A \rightarrow \infty$, so $\int_1^\infty \frac{1}{x^2} dx = 1$. **Converges.**
2. $\int_1^A x^{-1} dx = \ln(A) \rightarrow \infty$: **diverges.**
3. $\int_0^A e^{-x} dx = 1 - e^{-A} \rightarrow 1$, so $\int_0^\infty e^{-x} dx = 1$. **Converges.**
4. With $u = \ln x$: $\int_e^A \frac{1}{x(\ln x)^2} dx = [-1/\ln x]_e^A = 1 - 1/\ln(A) \rightarrow 1$, so the integral equals 1. **Converges.**
5. Not well defined. The function jumps from $-\infty$ to $+\infty$ near 0.

Leibniz Rule and Applications

Let

$$F(t) = \int_t^{t^2} \frac{x^2}{2} t \, dx.$$

Here the integrand $f(t, x) = \frac{tx^2}{2}$ depends on t , **and** both bounds $a(t) = t$, $b(t) = t^2$ depend on t . Assume $t > 1$.

1. Compute $F'(t)$ **without** computing the integral first.

Solution

Using the Leibniz Rule:

$$F'(t) = f(t, b(t)) b'(t) - f(t, a(t)) a'(t) + \int_{a(t)}^{b(t)} \frac{\partial f(t, x)}{\partial t} dx.$$

With $f(t, x) = \frac{tx^2}{2}$, $a(t) = t$ ($a' = 1$), $b(t) = t^2$ ($b' = 2t$), and $\partial f / \partial t = x^2/2$:

$$F'(t) = \frac{t(t^2)^2}{2} \cdot 2t - \frac{t t^2}{2} \cdot 1 + \int_t^{t^2} \frac{x^2}{2} dx = t^6 - \frac{t^3}{2} + \left[\frac{x^3}{6} \right]_t^{t^2} = t^6 - \frac{t^3}{2} + \frac{t^6 - t^3}{6} = \frac{7}{6}t^6 - \frac{2}{3}t^3.$$

2. Compute the explicit value of the integral, then differentiate it.

Solution

Factoring out t (constant in x):

$$F(t) = t \int_t^{t^2} \frac{x^2}{2} dx = t \left[\frac{x^3}{6} \right]_t^{t^2} = t \cdot \frac{t^6 - t^3}{6} = \frac{t^7}{6} - \frac{t^4}{6}.$$

3. Check that the two answers agree.

Solution

Differentiating $F(t) = \frac{t^7}{6} - \frac{t^4}{6}$:

$$F'(t) = \frac{7}{6}t^6 - \frac{4}{6}t^3 = \frac{7}{6}t^6 - \frac{2}{3}t^3,$$

which matches Exercise 1. ✓

Second-Price Auction

In this exercise, you will see how the Leibniz rule can be used in an economic context: the **second-price (Vickrey) auction**, where the highest bidder wins but pays the *second-highest* bid. We will show that bidding your true value is optimal and, remarkably, this holds *whatever* the other bidder does.

Bidder i has private value v_i (i.e. individual i is ready to pay up to v_i to acquire the good) and submits a bid $b_i \geq 0$. The opponent submits a bid B , which from i 's point of view is a random variable with density g and CDF G on $[0, \bar{b}]$ (think of $\bar{b}$ as the upper possible value for the good).

1. Explain why bidder i 's expected payoff from bidding b_i is

$$u_i(b_i) = \int_0^{b_i} (v_i - x) g(x) dx.$$

Solution

If the opponent's bid realises at x , then bidder i wins iff $x < b_i$; upon winning, i pays the second price x and nets $v_i - x$. Averaging over the opponent's bid (density g), and noting that losing yields 0,

$$u_i(b_i) = \int_0^{b_i} (v_i - x) g(x) dx + \int_{b_i}^{\bar{b}} 0 \cdot g(x) dx = \int_0^{b_i} (v_i - x) g(x) dx.$$

2. Differentiate u_i with respect to b_i (use Leibniz rule for a moving upper bound), and deduce the optimal bid.

Solution

To find the optimal bid, we take the derivative of the expected payoff with respect to b_i using the Leibniz integral rule.

Since the integrand $(v_i - x)g(x)$ does not depend on b_i , the partial derivative inside the integral is zero. We only evaluate the integrand at the moving upper bound, multiplied by the derivative of that bound (which is 1):

$$\frac{du_i}{db_i} = (v_i - b_i) g(b_i) \cdot \frac{d}{db_i}(b_i) + \int_0^{b_i} \frac{\partial}{\partial b_i} [(v_i - x)g(x)] dx = (v_i - b_i) g(b_i).$$

Because $g(x)$ is a probability density function, $g(b_i) \geq 0$. Therefore, the sign of the derivative depends entirely on the $(v_i - b_i)$ term:

- If $b_i < v_i$, then $\frac{du_i}{db_i} \geq 0$: raising the bid increases (or maintains) the expected payoff.
- If $b_i > v_i$, then $\frac{du_i}{db_i} \leq 0$: raising the bid decreases (or maintains) the expected payoff.

The expected payoff is therefore maximized when the derivative crosses from positive to negative, which occurs exactly at:

$$\boxed{b_i = v_i.}$$

Economic reading. Truthful bidding is optimal *regardless* of the opponent's bid distribution g . Because it never yields a strictly lower payoff than any other bid, it is a *weakly dominant* strategy. This is why second-price (Vickrey) auctions are called *strategy-proof*: bidders have no incentive to overthink or misreport their valuations, making the mechanism exceptionally clean to analyze.

Probability: Expectations and Variance

Discrete Random Variables

A **discrete random variable** X takes countably many values $x_1, x_2, \dots$ with **probability mass function** $p(x_i) = \Pr(X = x_i) \geq 0$ and $\sum_i p(x_i) = 1$. Its **expectation** and **variance** are

$$\mathbb{E}[X] = \sum_i x_i p(x_i), \quad \text{Var}(X) = \sum_i (x_i - \mathbb{E}[X])^2 p(x_i) = \mathbb{E}[X^2] - (\mathbb{E}[X])^2.$$

For any constants a, b : $\mathbb{E}[aX + b] = a \mathbb{E}[X] + b$ and $\text{Var}(aX + b) = a^2 \text{Var}(X)$.

A Discrete Lottery (warm-up)

You invest \$10 in a start-up. Your net payoff X (in dollars) is:

$$\Pr(X = -10) = 0.6, \quad \Pr(X = 10) = 0.3, \quad \Pr(X = 60) = 0.1,$$

i.e. the venture usually fails (you lose your \$10), sometimes returns \$10, and rarely delivers a big \$60 success.

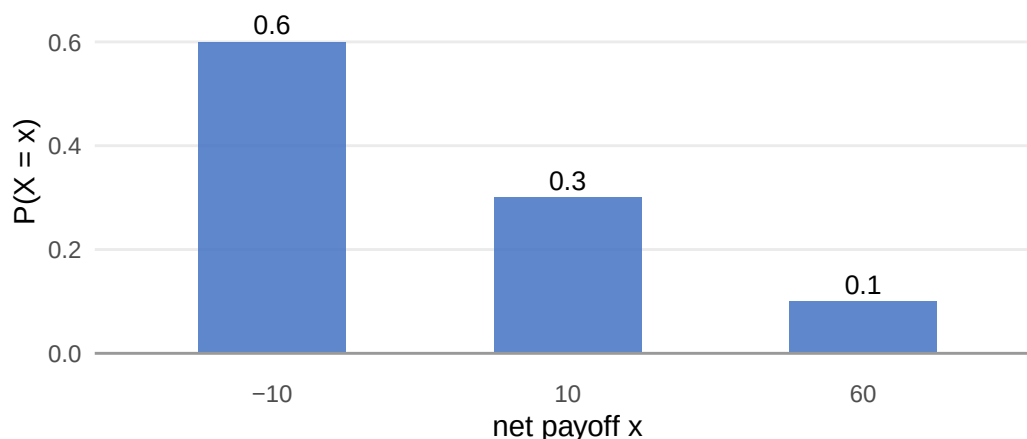


Figure 1

1. Check that this is a valid probability mass function.
2. Compute the expected net payoff $\mathbb{E}[X]$.
3. Compute $\text{Var}(X)$ and the standard deviation σ .
4. A fund lets you take **twice** (\$20 in the startup) the position for a flat \$2 fee, so your payoff becomes $Y = 2X - 20$. Use the linearity rules to find $\mathbb{E}[Y]$, $\text{Var}(Y)$ and σ_Y .
5. A safe government bond returns \$13 for sure. Compare its mean and variance with those of X . Which would a **risk-averse** investor prefer, and why?

Solution

1. $0.6 + 0.3 + 0.1 = 1$ and all probabilities are ≥ 0 . Valid. ✓

2.

$$\mathbb{E}[X] = (-10)(0.6) + (10)(0.3) + (60)(0.1) = -6 + 3 + 6 = 3.$$

On average the investment returns \$3.

3.

$$\mathbb{E}[X^2] = (100)(0.6) + (100)(0.3) + (3600)(0.1) = 60 + 30 + 360 = 450,$$

$$\text{Var}(X) = 450 - 3^2 = 441, \quad \sigma = \sqrt{441} = 21.$$

4. By linearity, with $a = 2$, $b = -2$:

$$\mathbb{E}[Y] = 2\mathbb{E}[X] - 2 = 2(3) - 2 = 4, \quad \text{Var}(Y) = 2^2 \text{Var}(X) = 4(441) = 1764, \quad \sigma_Y = 42.$$

(The fee shifts the mean but not the spread; doubling scales the standard deviation by 2.)

5. The bond has mean 3 (*same* as X) but variance 0. The start-up has the same expected return with $\sigma = 21$. A risk-averse investor prefers the bond: identical expected payoff, no risk. Equivalently, to attract them into the start-up you would need a higher expected return — a **risk premium**.

Continuous Random Variables

A **continuous random variable** X is described by a **probability density function** (PDF) $f : \mathbb{R} \rightarrow \mathbb{R}_+$ satisfying:

$$\int_{-\infty}^{+\infty} f(x) dx = 1.$$

The probability that X falls in an interval $[a, b]$ is:

$$\Pr(a \leq X \leq b) = \int_a^b f(x) dx.$$

The **cumulative distribution function** (CDF) is defined as:

$$F(t) = \Pr(X \leq t) = \int_{-\infty}^t f(x) dx.$$

The **expectation** (mean) of X is:

$$\mathbb{E}[X] = \int_{-\infty}^{+\infty} x f(x) dx.$$

The **variance** of X measures the spread around the mean:

$$\text{Var}(X) = \mathbb{E}[(X - \mathbb{E}[X])^2] = \int_{-\infty}^{+\infty} (x - \mathbb{E}[X])^2 f(x) dx = \mathbb{E}[X^2] - (\mathbb{E}[X])^2.$$

Waiting Times at a Service Desk

A firm operates a customer service desk. The waiting time T (in minutes) before a customer is served is modelled as a continuous random variable with probability density function:

$$f(t) = \lambda e^{-\lambda t}, \quad t \geq 0,$$

where $\lambda > 0$ is a parameter capturing how fast the desk operates (a higher λ means shorter average wait). This is the **exponential distribution** with parameter λ .

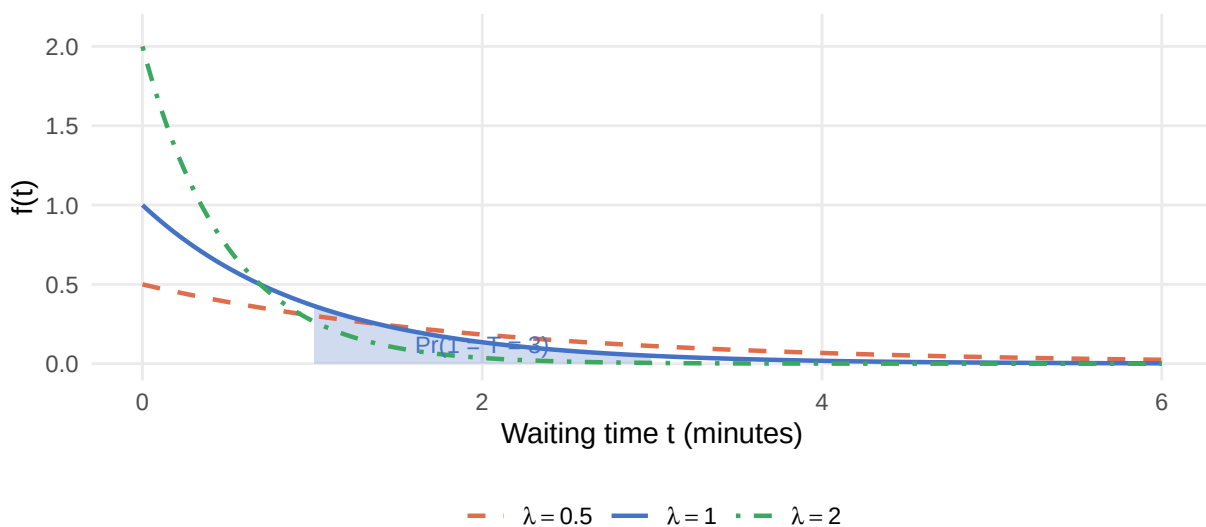


Figure 2

1. Verify that f is a valid PDF, i.e. show that $\int_0^{+\infty} \lambda e^{-\lambda t} dt = 1$ and it's never negative.
2. Compute the CDF $F(t) = \Pr(T \leq t)$.
3. A customer is “well served” if they wait less than 3 minutes. A manager sets $\lambda = 1$. What is the probability that a customer waits more than 3 minutes?
4. Compute $\mathbb{E}[T]$ and interpret it in terms of the firm's service desk.
5. Compute $\text{Var}(T)$ using $\text{Var}(T) = \mathbb{E}[T^2] - (\mathbb{E}[T])^2$. (*Hint: use integration by parts*)

Solution

1. We compute:

$$\int_0^{+\infty} \lambda e^{-\lambda t} dt = \lambda \left[-\frac{1}{\lambda} e^{-\lambda t} \right]_0^{+\infty} = \lambda \cdot \frac{1}{\lambda} = 1. \quad \checkmark$$

2. For $t \geq 0$:

$$F(t) = \int_0^t \lambda e^{-\lambda s} ds = [-e^{-\lambda s}]_0^t = 1 - e^{-\lambda t}.$$

3. With $\lambda = 1$:

$$\Pr(T > 3) = 1 - F(3) = e^{-3} \approx 0.050.$$

About 5% of customers wait more than 3 minutes.

4. Integration by parts with $u = t$, $dv = \lambda e^{-\lambda t} dt$, $v = -e^{-\lambda t}$:

$$\mathbb{E}[T] = \int_0^{+\infty} t \lambda e^{-\lambda t} dt = [-te^{-\lambda t}]_0^{+\infty} + \int_0^{+\infty} e^{-\lambda t} dt = 0 + \frac{1}{\lambda} = \frac{1}{\lambda}.$$

On average, customers wait $1/\lambda$ minutes. A faster desk (larger λ) reduces the mean wait.

5. First, $\mathbb{E}[T^2] = \int_0^{\infty} t^2 \lambda e^{-\lambda t} dt$. Integrating by parts ($u = t^2$, $dv = \lambda e^{-\lambda t} dt$):

$$\mathbb{E}[T^2] = [-t^2 e^{-\lambda t}]_0^{\infty} + \int_0^{\infty} 2t e^{-\lambda t} dt = 0 + \frac{2}{\lambda} \int_0^{\infty} t \lambda e^{-\lambda t} dt = \frac{2}{\lambda} \cdot \frac{1}{\lambda} = \frac{2}{\lambda^2}.$$

Therefore

$$\text{Var}(T) = \frac{2}{\lambda^2} - \frac{1}{\lambda^2} = \frac{1}{\lambda^2}.$$

The standard deviation $1/\lambda$ equals the mean: the exponential is highly dispersed relative to its mean.

General economic applications

Consumer and Producer Surplus

Consumer and Producer Surplus

Consider a market with inverse demand $P_D(q)$ and inverse supply $P_S(q)$, and let q^* be the equilibrium quantity at price p^* .

- The **consumer surplus** is the aggregate gain to buyers:

$$CS = \int_0^{q^*} (P_D(q) - p^*) dq.$$

- The **producer surplus** is the aggregate gain to sellers:

$$PS = \int_0^{q^*} (p^* - P_S(q)) dq.$$

Consider a market for a consumption good with the following inverse demand and supply curves:

$$P_D(q) = 12 - q^2, \quad P_S(q) = 2q + 2.$$

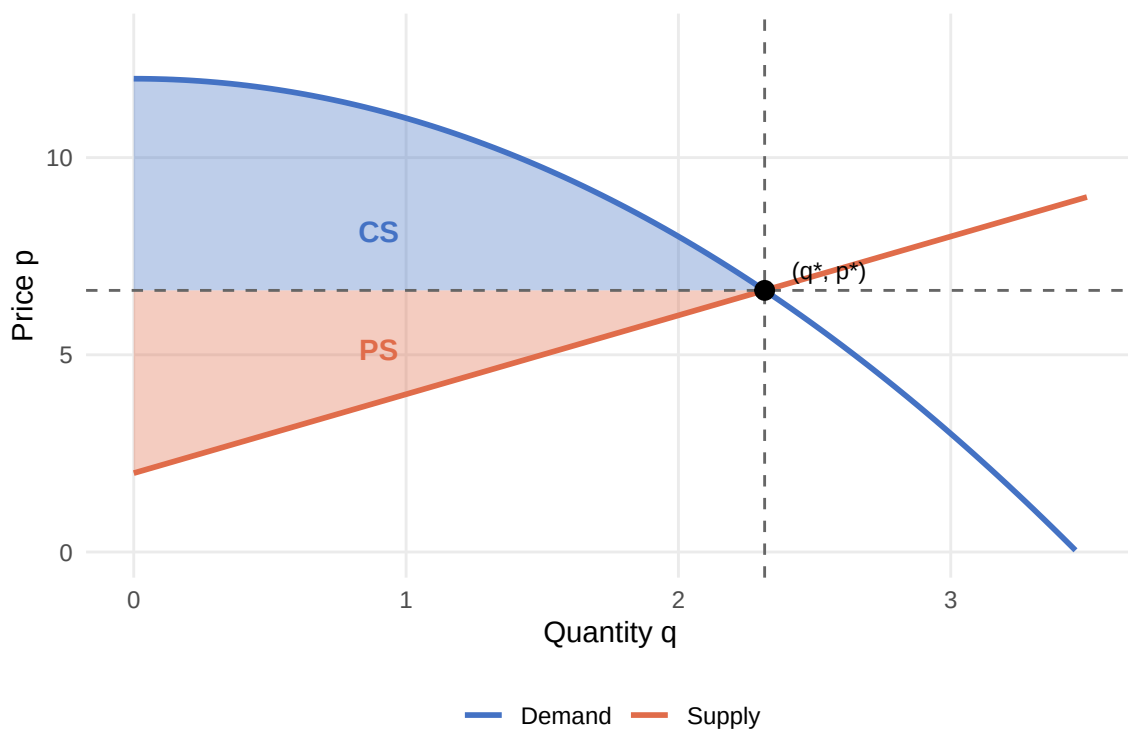


Figure 3

1. Find the equilibrium price p^* and quantity q^* (*hint: supply must equal demand*).
2. Compute the consumer surplus CS .
3. Compute the producer surplus PS .
4. What is the total surplus $TS = CS + PS$? Briefly interpret this number.

Solution

1. Setting $P_D(q) = P_S(q)$:

$$12 - q^2 = 2q + 2 \implies q^2 + 2q - 10 = 0 \implies q^* = -1 + \sqrt{11} \approx 2.317.$$

Then $p^* = 2q^* + 2 = 2\sqrt{11} \approx 6.633$ (equivalently $p^* = 12 - (q^*)^2$).

2. Substituting $p^* = 12 - (q^*)^2$, the integrand becomes $(12 - q^2) - (12 - (q^*)^2) = (q^*)^2 - q^2$, so

$$CS = \int_0^{q^*} ((q^*)^2 - q^2) dq = \left[(q^*)^2 q - \frac{q^3}{3} \right]_0^{q^*} = (q^*)^3 - \frac{(q^*)^3}{3} = \frac{2}{3}(q^*)^3.$$

With $q^* = \sqrt{11} - 1$ we have $(q^*)^3 = 14\sqrt{11} - 34$, hence

$$CS = \frac{2}{3} (14\sqrt{11} - 34) = \frac{28\sqrt{11}-68}{3} \approx 8.29.$$

3. Since $p^* - 2 = 2q^*$,

$$PS = \int_0^{q^*} (p^* - 2 - 2q) dq = (p^* - 2)q^* - (q^*)^2 = 2(q^*)^2 - (q^*)^2 = (q^*)^2.$$

So $PS = (\sqrt{11} - 1)^2 = 12 - 2\sqrt{11} \approx 5.37$.

4. $TS = CS + PS = \frac{28\sqrt{11}-68}{3} + 12 - 2\sqrt{11} = \frac{22\sqrt{11}-32}{3} \approx 13.66$. This is the total value created by the market: the aggregate gains from trade enjoyed by all buyers and sellers at the competitive equilibrium.

Lorenz Curve and Inequality

Lorenz Curve and Gini Coefficient

Let F be the CDF of income in a population, with mean μ . The **Lorenz curve** $L : [0, 1] \rightarrow [0, 1]$ maps the cumulative population share p to the corresponding cumulative income share:

$$L(p) = \frac{1}{\mu} \int_0^{F^{-1}(p)} x f(x) dx.$$

It satisfies $L(0) = 0$, $L(1) = 1$, and $L(p) \leq p$ for all p (the poorest p fraction always earns less than p fraction of total income). The **Gini coefficient** measures the area between the line of perfect equality and the Lorenz curve:

$$G = 1 - 2 \int_0^1 L(p) dp.$$

$G = 0$ means perfect equality; $G = 1$ means one person earns everything.

This exercise should challenge you, and take you some time. Do not panic, proceed step-by-step.

Suppose income y follows a **power-law distribution** on $[1, e]$ with PDF

$$f(y) = \frac{\alpha y^{\alpha-1}}{e^\alpha - 1}, \quad y \in [1, e], \quad \alpha > 0,$$

where $y = 1$ is the minimum income and $y = e$ the maximum.

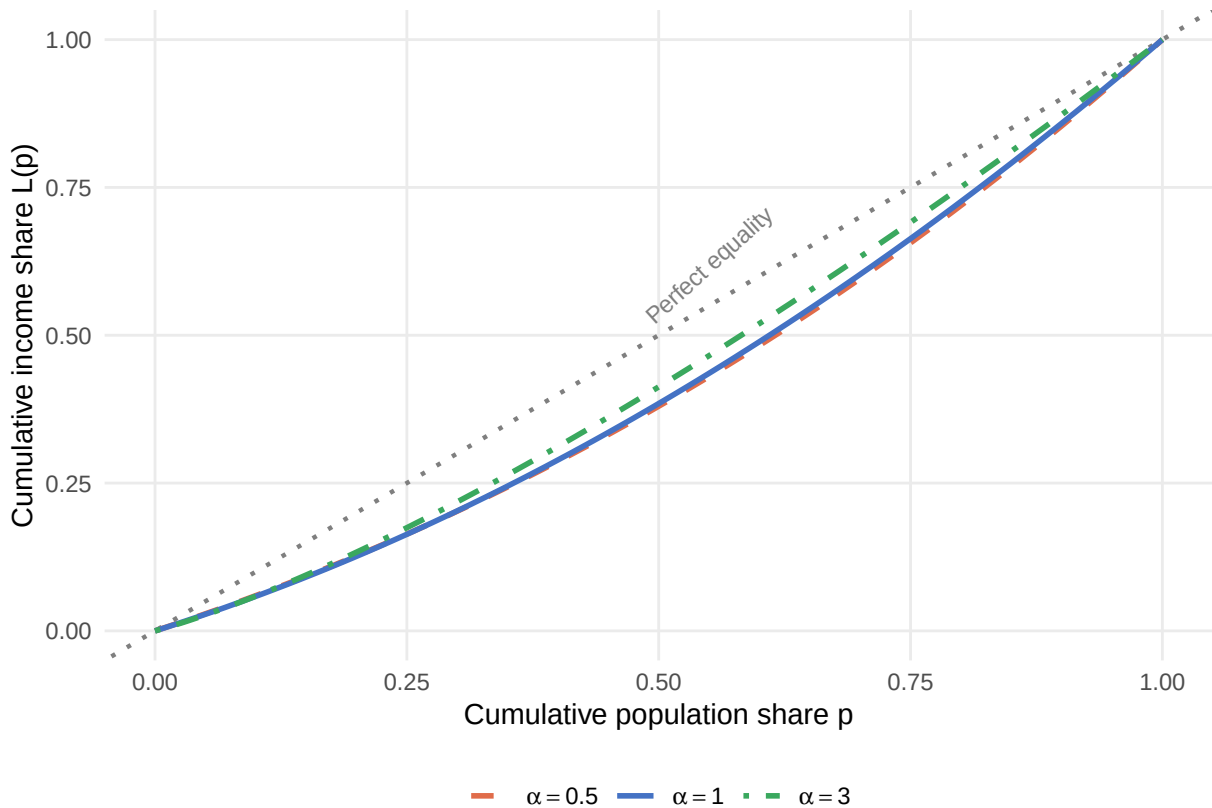


Figure 4

1. Verify that f is a valid PDF on $[1, e]$.
2. Compute the mean income $\mu = \mathbb{E}[Y]$.

3. Compute the CDF $F(y)$, then invert it to get the quantile function $F^{-1}(p)$. *Example: to invert the function $f(x) = x^2 + 1$, you start with $y = x^2 + 1$, then try to express x as a function of y instead of the reverse $\rightarrow y - 1 = x^2$ and finally $x = \sqrt{y - 1}$ which is defined only for $y \geq 1$.*
4. Show that the Lorenz curve is

$$L(p) = \frac{(p(e^\alpha - 1) + 1)^{(\alpha+1)/\alpha} - 1}{e^{\alpha+1} - 1}.$$

Hint: proceed step-by-step. First, replace $q \equiv F^{-1}(p)$ and take only the integral. Then divide by μ , and only then input $F^{-1}(p)$.

5. What happens to the Lorenz curve as $\alpha \rightarrow \infty$? What does this imply economically? (Use the figure to guide your intuition.)
6. Compute the Gini coefficient $G = 1 - 2 \int_0^1 L(p) dp$ for $\alpha = 1$.

Note that we worked with a function defined on $[1, e]$ for convenience. But, more generally, you can use the following distributional form to model inequalities:

$$f(y) = \frac{\alpha y^{\alpha-1}}{M^\alpha - 1}, \quad y \in [1, M], \quad \alpha > 0,$$

where $M > 1$ and $\alpha > 0$ are parameters that govern respectively the income inequality (M represents how much the top makes compared to the bottom of the income distribution) and the curvature of the distribution (income inequalities).

Solution

1.

$$\int_1^e \frac{\alpha y^{\alpha-1}}{e^\alpha - 1} dy = \frac{1}{e^\alpha - 1} [y^\alpha]_1^e = \frac{e^\alpha - 1}{e^\alpha - 1} = 1. \quad \checkmark$$

2.

$$\mu = \int_1^e y \cdot \frac{\alpha y^{\alpha-1}}{e^\alpha - 1} dy = \frac{\alpha}{e^\alpha - 1} \cdot \frac{e^{\alpha+1} - 1}{\alpha + 1} = \frac{\alpha(e^{\alpha+1} - 1)}{(\alpha + 1)(e^\alpha - 1)}.$$

3.

$$F(y) = \int_1^y \frac{\alpha t^{\alpha-1}}{e^\alpha - 1} dt = \frac{y^\alpha - 1}{e^\alpha - 1}.$$

Solving $p = (y^\alpha - 1)/(e^\alpha - 1)$ for y :

$$F^{-1}(p) = (p(e^\alpha - 1) + 1)^{1/\alpha}.$$

4. With $q = F^{-1}(p)$,

$$\int_1^q y f(y) dy = \frac{\alpha}{e^\alpha - 1} \cdot \frac{q^{\alpha+1} - 1}{\alpha + 1}.$$

Dividing by $\mu = \frac{\alpha}{e^\alpha - 1} \cdot \frac{e^{\alpha+1} - 1}{\alpha + 1}$, the factor $\frac{\alpha}{(e^\alpha - 1)(\alpha + 1)}$ cancels entirely:

$$L(p) = \frac{q^{\alpha+1} - 1}{e^{\alpha+1} - 1} = \frac{(p(e^\alpha - 1) + 1)^{(\alpha+1)/\alpha} - 1}{e^{\alpha+1} - 1}.$$

Check: at $p = 1$, $p(e^\alpha - 1) + 1 = e^\alpha$, so the numerator is $(e^\alpha)^{(\alpha+1)/\alpha} - 1 = e^{\alpha+1} - 1$ and $L(1) = 1$. $\checkmark$

5. As $\alpha \rightarrow \infty$, mass concentrates near $y = e$ (the top income), so everyone earns almost the same and $L(p) \rightarrow p$: income becomes **perfectly equal**. A larger α therefore *reduces* inequality (the curve moves toward the diagonal, as in the figure).

6. For $\alpha = 1$ the density is $f(y) = \frac{1}{e-1}$ — i.e. *uniform* on $[1, e]$ — and $(\alpha + 1)/\alpha = 2$, so

$$L(p) = \frac{(p(e-1) + 1)^2 - 1}{e^2 - 1}.$$

With the substitution $u = p(e-1) + 1$ ($du = (e-1)dp$, $u : 1 \rightarrow e$),

$$\int_0^1 (p(e-1) + 1)^2 dp = \frac{1}{e-1} \cdot \frac{e^3 - 1}{3} = \frac{e^2 + e + 1}{3},$$

so

$$\int_0^1 L(p) dp = \frac{1}{e^2 - 1} \left(\frac{e^2 + e + 1}{3} - 1 \right) = \frac{e^2 + e - 2}{3(e^2 - 1)} = \frac{e + 2}{3(e + 1)}.$$

Therefore

$$G = 1 - 2 \cdot \frac{e + 2}{3(e + 1)} = \frac{e - 1}{3(e + 1)} \approx 0.154.$$

(Consistency check: for a uniform distribution on $[a, b]$ the Gini is $\frac{b-a}{3(a+b)}$, which here gives $\frac{e-1}{3(e+1)}$ — exactly our result.)

Wage Bargaining

A firm and a worker negotiate over a wage w . The job generates a surplus S (the total value created if they reach an agreement), but this surplus is **uncertain**: it is drawn from a uniform distribution on $[0, \bar{s}]$, so its PDF is $f(s) = \frac{1}{\bar{s}} \mathbf{1}(0 \leq s \leq \bar{s})$.

The worker has an outside option $\bar{u} > 0$ (e.g. unemployment benefits): they will only accept the job if $w \geq \bar{u}$. The firm will only offer the job if $S - w \geq 0$. Hence a deal is struck **only when** $S \geq \bar{u}$.

When a deal is struck, the parties solve the **Nash bargaining problem**: they choose w to maximise the Nash product

$$N(w, S) = (w - \bar{u})^\beta \cdot (S - w)^{1-\beta},$$

where $\beta \in (0, 1)$ is the worker's bargaining power.



Figure 5

1. Taking S as given (a known constant), find the wage $w^*(S)$ that maximises $N(w, S)$. (Hint: maximise $\ln N$ by solving $\frac{\partial \ln N}{\partial w} = 0$.)
2. Explain why the expected negotiated wage, conditional on a deal, is

$$\mathbb{E}[w^* \mid S \geq \bar{u}] = \frac{\int_{\bar{u}}^{\bar{s}} w^*(s) f(s) ds}{\Pr(S \geq \bar{u})},$$

and compute it explicitly.

3. How does the expected wage change with β ? With $\bar{u}$? Give the economic intuition.
4. Compute the expected **firm profit** conditional on a deal, $\mathbb{E}[S - w^* \mid S \geq \bar{u}]$.

Solution

1. Taking logs: $\ln N = \beta \ln(w - \bar{u}) + (1 - \beta) \ln(S - w)$. Differentiating in w :

$$\frac{\partial \ln N}{\partial w} = \frac{\beta}{w - \bar{u}} - \frac{1 - \beta}{S - w} = 0 \implies \beta(S - w) = (1 - \beta)(w - \bar{u}).$$

Solving,

$$w^*(S) = \beta S + (1 - \beta)\bar{u}.$$

The worker earns a weighted average of the surplus and their outside option, with weight β on the surplus.

2. Since $S \sim U[0, \bar{s}]$, $\Pr(S \geq \bar{u}) = (\bar{s} - \bar{u})/\bar{s}$ and $f(s) = 1/\bar{s}$. Thus

$$\begin{aligned} \mathbb{E}[w^* \mid S \geq \bar{u}] &= \frac{\int_{\bar{u}}^{\bar{s}} (\beta s + (1 - \beta)\bar{u}) \frac{1}{\bar{s}} ds}{(\bar{s} - \bar{u})/\bar{s}} = \frac{1}{\bar{s} - \bar{u}} \int_{\bar{u}}^{\bar{s}} (\beta s + (1 - \beta)\bar{u}) ds \\ &= \frac{1}{\bar{s} - \bar{u}} \left[\frac{\beta s^2}{2} + (1 - \beta)\bar{u}s \right]_{\bar{u}}^{\bar{s}} = \beta \frac{\bar{s} + \bar{u}}{2} + (1 - \beta)\bar{u} = \bar{u} + \beta \frac{\bar{s} - \bar{u}}{2}. \end{aligned}$$

3. The expected wage $\bar{u} + \beta(\bar{s} - \bar{u})/2$ is **increasing in** β (more bargaining power raises the wage) and **increasing in** $\bar{u}$ (a better outside option strengthens the worker's threat point, pushing wages up).
4. We have $S - w^*(S) = (1 - \beta)(S - \bar{u})$. Moreover, the probability conditional on a deal:

$$f_{S|S \geq \bar{u}} = \frac{f_{S \wedge S \geq \bar{u}}}{P(S \geq \bar{u})} = \frac{\frac{1}{\bar{s}} \mathbf{1}(\bar{u} \leq s \leq \bar{s})}{\frac{\bar{s} - \bar{u}}{\bar{s}}} = \frac{1}{\bar{s} - \bar{u}}, \quad s \in [\bar{u}, \bar{s}].$$

You discovered an interesting thing! Truncating a uniform gives a uniform on the truncated interval.

Now, you may compute the expected payoff:

$$\mathbb{E}(S - w^*(S)) = (1 - \beta)\mathbb{E}((S - \bar{u})) = (1 - \beta) \int_{\bar{u}}^{\bar{s}} x \frac{1}{\bar{s} - \bar{u}} dx = (1 - \beta) \frac{\bar{s} - \bar{u}}{2}$$

$$S \sim U[\bar{u}, \bar{s}] \text{ so } \mathbb{E}[S - \bar{u} \mid S \geq \bar{u}] = (\bar{s} - \bar{u})/2$$

$$\mathbb{E}[S - w^* \mid S \geq \bar{u}] = (1 - \beta) \frac{\bar{s} - \bar{u}}{2}.$$

This is **decreasing in** β : stronger worker bargaining power compresses firm profits.